Fourth Semester B. Tech. (Electronics and Communication Engineering) Examination

ANALOG CIRCUITS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory.
(2) All questions carry marks as indicated against them.
(3) Assume suitable data and illustrate answer with sketches wherever necessary.

1. (a) Develop the expression for input and output resistance of current series feedback amplifier. 8(CO1,2)

   (b) Explain sampling and mixing network. 2(CO1,2)

2. (a) Evaluate the efficiency of class A amplifier. 5(CO3)

   (b) What is a cross-over distortion ? How crossover distortion is eliminated? 5(CO3)

3. (a) Explain the working of Wein bridge oscillator. Develop the formula for the frequency of oscillations. 8(CO2,3)

   (b) Evaluate the frequency of the oscillations of a transtorized Colpitts oscillator having C1 = 150 pf, C2 = 1.5 nf and L = 50 μH. 2(CO2,3)

4. (a) For current miror circuit prove that the output current is equal to the input current. 5(CO3)

   (b) Design dual input balanced output differential amplifier using diode current bias to meet the following specifications :

   (1) Supply voltages Vs = ±12 V

(2) Emitter current I_E in each differential amplifier 1.5 mA.

(3) Voltage gain = 60. 5(CO3)

5. (a) Derive the equation for a full wave rectifier using op-amp. Draw the circuit diagram explain its working. 5(CO4)

(b) What is a Differentiator ? What are its limitations for ideal differentiator? How they are overcome in practical differentiator ? 5(CO4)

6. (a) Design 2nd order Butterworth Low pass filter at a high cut-off frequency 1 KHz. 4(CO5)

(b) Explain briefly working of IC 555. 6(CO5)